



鋰離子電池組健康狀態估算方法之回顧

A review of state-of-health estimation for lithium-ion
battery packs

台灣科技大學 AI所

指導教授：羅一峰 博士

學生：林志龍



Outline

- Introduction
- Degradation Mechanisms
- Health Feature Extraction
- SOH Estimation Models
- Comparison of Models
- Challenges & Future Prospects
- Conclusion



Introduction

Why SOH Matters:

- SOH is the cornerstone of **safety and reliability** for Electric Vehicles (EVs) and Energy Storage Systems.

Why Focus on "Battery Packs" ?

- **Real-world Application:** EVs operate on battery *packs* (series/parallel), **not isolated cells**
- **The "Barrel Effect":** Pack performance is limited by the weakest cell. Inconsistency between cells accelerates degradation faster than single-cell aging .

The Research Gap:

- **Limited Scope:** Most existing reviews focus solely on **single-cell estimation**.
- **The Missing Link:** There is a scarcity of comprehensive reviews addressing pack-level SOH estimation and the impact of cell inconsistency.



Degradation Mechanisms

How Lithium-ion Batteries Work

Core Mechanism:

- Relies on the reversible migration of Lithium ions (Li^+) between the cathode and anode.
- **Charging:** Li^+ moves from **cathode** $\rightarrow$ **anode** (Lithium enrichment at anode)
- **Discharging:** Li^+ moves from **anode** $\rightarrow$ **cathode** (Lithium depletion at anode)

Key Components:

- **Electrodes:** **Positive** (e.g., LiCoO_2) and **Negative** (Carbon materials)
- **Separator:** Physically isolates electrodes while allowing ion flow.
- **Electrolyte:** Medium for ion transport.

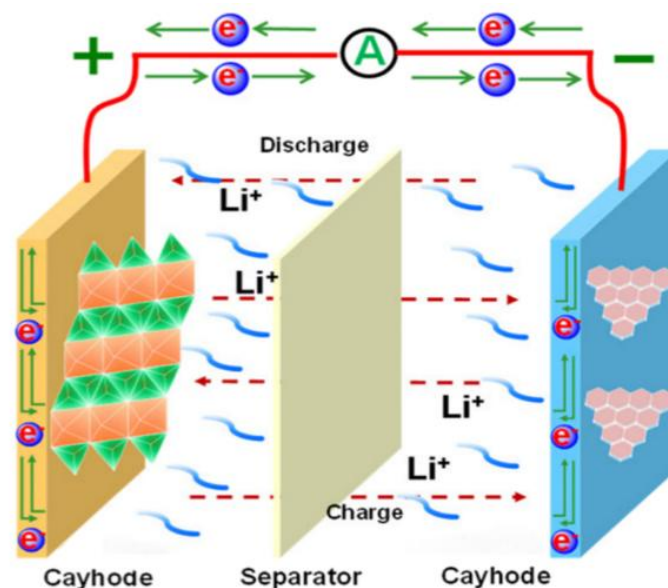


Fig. 1. Lithium-ion battery working diagram.



Degradation Mechanisms

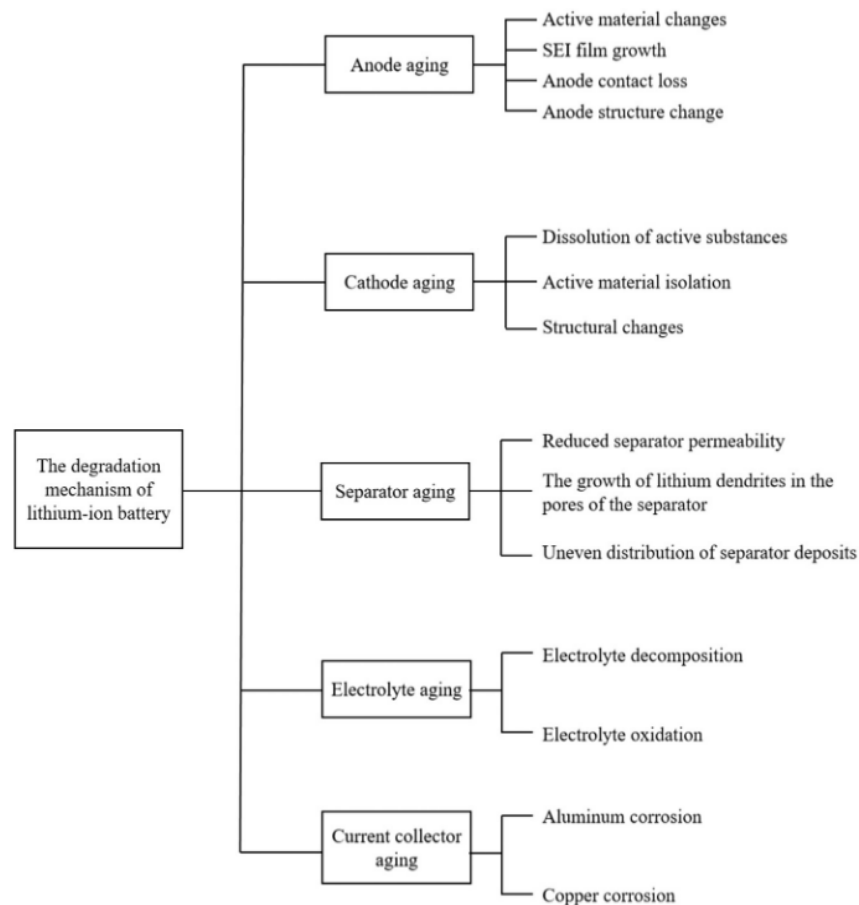
Cell-Level Degradation Mechanisms

Chemical Degradation:

- **SEI Growth:** Continuous formation of Solid Electrolyte Interphase consumes active lithium and increases **internal resistance**.
- **Electrolyte Decomposition:** **Reduction/oxidation** reactions alter electrolyte composition.
- **Active Material Loss:** Dissolution of electrode structures reduces **reversible capacity**.

Mechanical Degradation:

- **Volume Changes:** Stress from particle expansion/contraction during cycling leads to **structural damage**.





Degradation Mechanisms

Impact of Fast Charging on Battery Health

Thermal Stress :

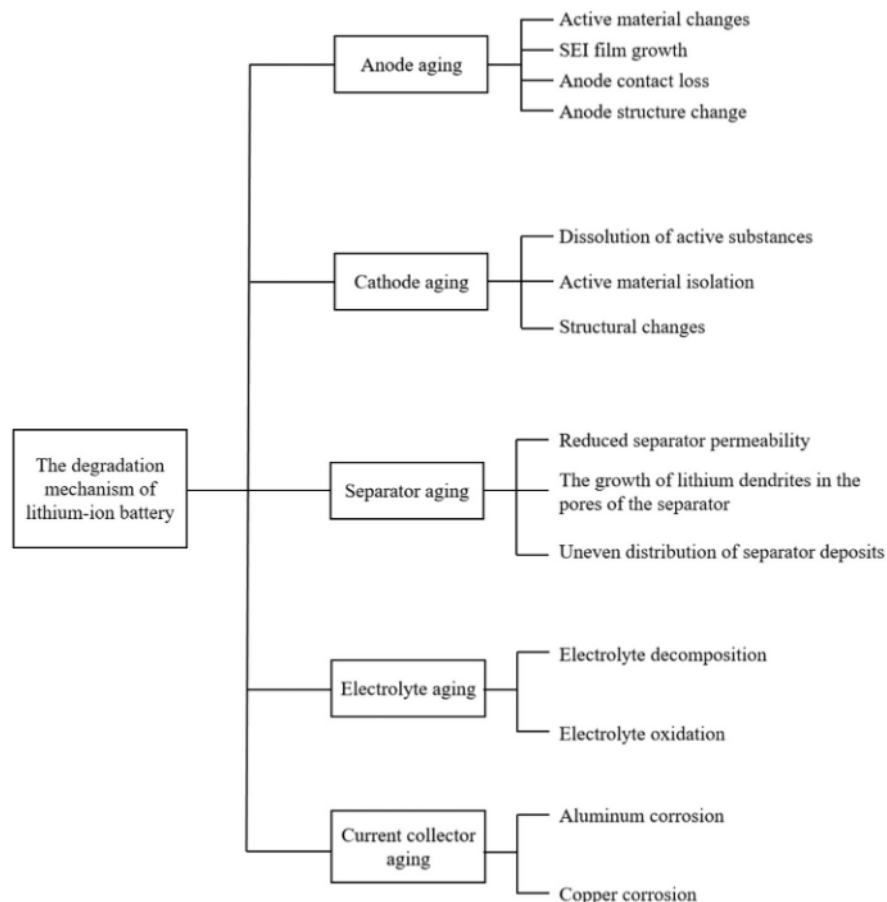
- **High currents** elevate internal temperature, accelerating chemical reactions and electrolyte decomposition

Lithium Plating (Major Risk) :

- Rapid charging causes Li^+ deposition on the anode surface instead of intercalation.
- Leads to **Lithium Dendrite** formation, causing capacity fade and safety risks (**short circuits**)

Low-Temperature Synergy :

- Fast charging at low temps (e.g., -5°) significantly **increases plating area** and **internal resistance**





Degradation Mechanisms

Degradation Mechanism of Battery Packs

The "Barrel Effect" :

- In series connections, the **weakest cell** determines the entire pack's usable capacity and performance

The "Vicious Cycle" of Inconsistency:

- **Initial State:** Minor manufacturing differences lead to uneven heat and current distribution.
- **Operation:** Variations in internal resistance cause uneven aging rates across cells
- **Result:** Inconsistency expands over time, accelerating overall pack failure

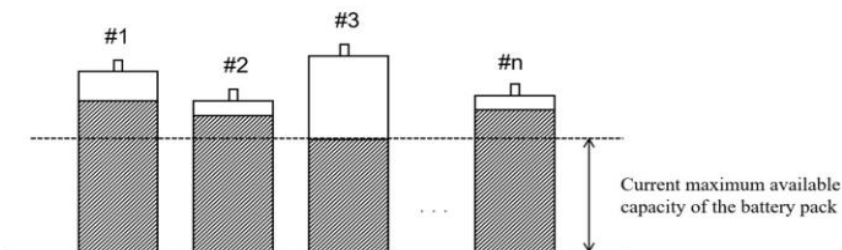


Fig. 3. Lithium-ion battery pack inconsistency.

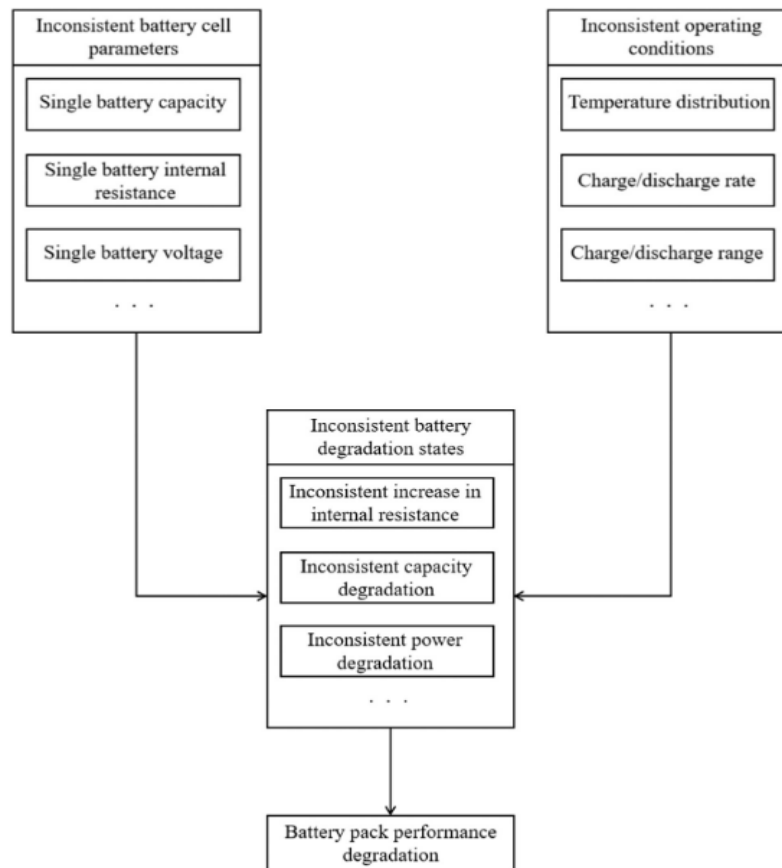


Fig. 4. Degradation mechanisms of lithium-ion battery packs.



Health Feature Extraction

Extracting Features from Charging Curves

Concept :

- Extracts "Health Indicators" (HIs) from standard voltage and current data
- Uses **Incremental Capacity (IC) curves** (dQ/dV) to visualize internal electrochemical changes

Mechanism:

- Peaks on the **IC curve** correspond to specific electrochemical reaction stages.
- **Aging Signature:** As the battery ages, **IC peaks shift** in position and decrease in amplitude

Challenges:

- Highly sensitive to **measurement noise**; requires smoothing (e.g., Gaussian filters) for accuracy

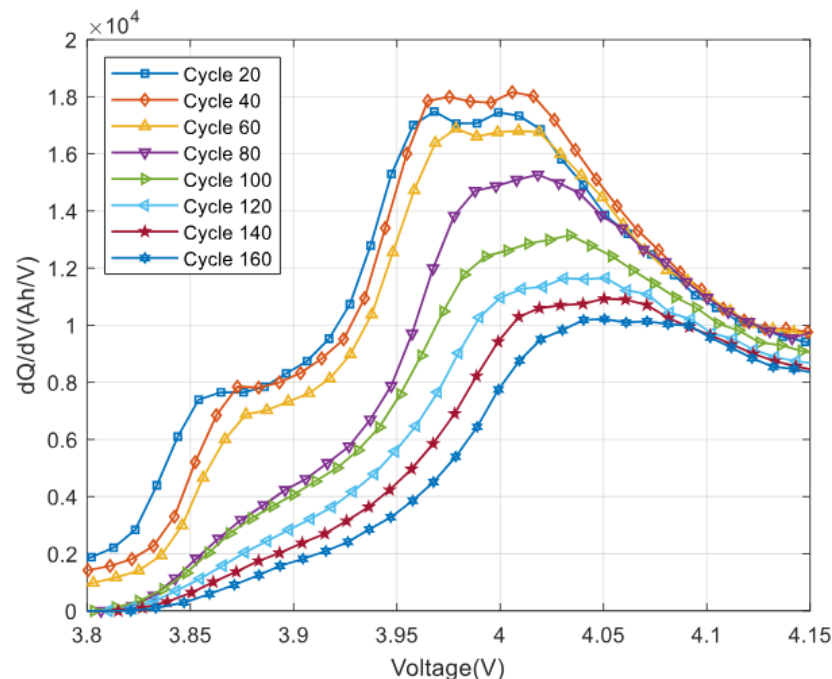


Fig. 5. Variation of IC curve of lithium-ion batteries at different cycles.



Health Feature Extraction

Time-Domain & Frequency-Domain Features

Time-Based Features (Simple & Fast) :

- Monitors the **duration** of charging phases.
- **Trend:** In CC-CV charging, the Constant Voltage (CV) time *increases* while Constant Current (CC) time *decreases* with aging
- **Pros:** Low computational cost
Cons: Requires strict charging protocols

Frequency-Domain (EIS) (Deep Analysis):

- Uses **Electrochemical Impedance Spectroscopy (EIS)** to measure internal resistance and SEI film growth
- **Pros:** Provides deep insight into internal kinetics
Cons: Equipment is expensive and time-consuming

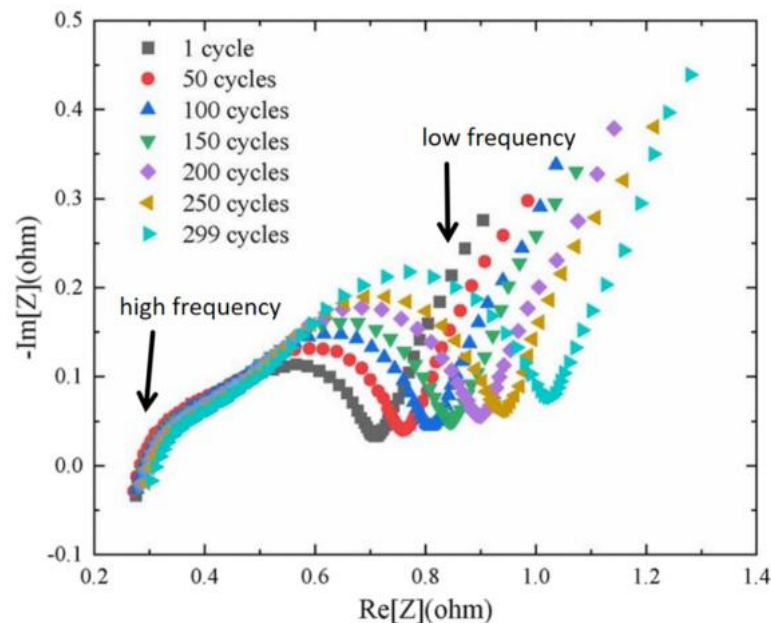


Fig. 6. Variation of EIS data of lithium-ion batteries at different charging and discharging cycles.



Health Feature Extraction

Pack-Level Features & Comparison

Table 1

Comparison of different degradation feature recognition methods.

Method	Scope of application	Advantages	Limitations
Incremental capacity analysis-based health feature	Lithium-ion battery Lithium-ion battery packs	- Based on measurable voltage and current parameters, degradation feature extraction can be achieved; - It can be used simultaneously to analyze the degradation mechanism of lithium-ion battery monomers	- Highly sensitive to noise in voltage and current parameter measurements - Low-rate charging and discharging of battery or battery pack
Time-based health feature	Lithium-ion battery Lithium-ion battery packs	- Based on measurable voltage and current parameters, degradation feature extraction can be achieved - Effective degradation features can be extracted throughout the entire process of both the battery cell and the battery pack	- The extraction of degradation features during the discharge stage requires a constant discharge current, which is difficult to meet in practical applications - Some parameters require control of the battery's status
Frequency domain analysis-based health feature	Lithium-ion battery Lithium-ion battery packs	- It is capable of non-invasively characterizing changes in electrochemical kinetic processes within a battery - Information-rich and highly sensitive	- High equipment costs and time-consuming measurements - High requirements for data processing and analysis
Multi-physical sensing technologies-based health feature	Lithium-ion battery Lithium-ion battery packs	- Complementary information and strong robustness - Comprehensive data and real-time monitoring of batteries	- System complexity and difficulty in data processing - Higher cost and lower reliability due to use of multiple sensors
Battery pack inconsistency-based health feature	Lithium-ion battery packs	- It can directly characterize the performance degradation state of battery packs and respond to the performance degradation mechanism of battery packs - The analysis results are beneficial for continuing the individual equilibrium within the battery pack	- Evaluation of inconsistent available parameters is difficult to identify online - There is a lack of effective methods for the convergence of inconsistency evaluation results



SOH Estimation Models

Model-Based Approaches (The "White Box")

Core Concept:

- Estimates internal states (SOC/SOH) by **observing external measurements** (Voltage, Current, Temp).
- Uses **State-Space Models** to mathematically represent battery physics.

Key Algorithms :

- Kalman Filter (KF)**: Recursive algorithm suitable for **linear systems**. Fast and low complexity.
- Extended Kalman Filter (EKF)**: Modified version to handle **non-linear** battery behaviors.
- Particle Filter (PF)**: Uses Monte Carlo sampling; excellent for **non-Gaussian noise** and **highly non-linear systems**.

Pros & Cons:

- Pro**: Real-time capability, low computational cost.
- Con**: Heavily relies on the accuracy of the initial battery model.



SOH Estimation Models

Data-Driven Approaches (The "Black Box")

Core Concept :

- Treats the battery as a **"Black Box"**. No need to understand complex internal chemical reactions .
- **Learns mapping rules** between "Health Features" and SOH directly from historical data.

Architectures:

- **SVM (Support Vector Machine)**: Identifies an optimal hyperplane to **separate data**; highly effective for small datasets .
- **RVM (Relevance Vector Machine)**: A Bayesian sparse model that improves on SVM by providing **probabilistic predictions (uncertainty)** .
- **GPR (Gaussian Process Regression)**: A non-parametric model that uniquely outputs **confidence intervals (reliability)** alongside the prediction .
- **NN (Neural Network)**: **Mimics human neurons to excel** at processing complex, non-linear data without detailed physical knowledge .
- **ELM (Extreme Learning Machine)**: Uses **random weights for single-layer networks**, achieving superior training speed compared to traditional NNs .

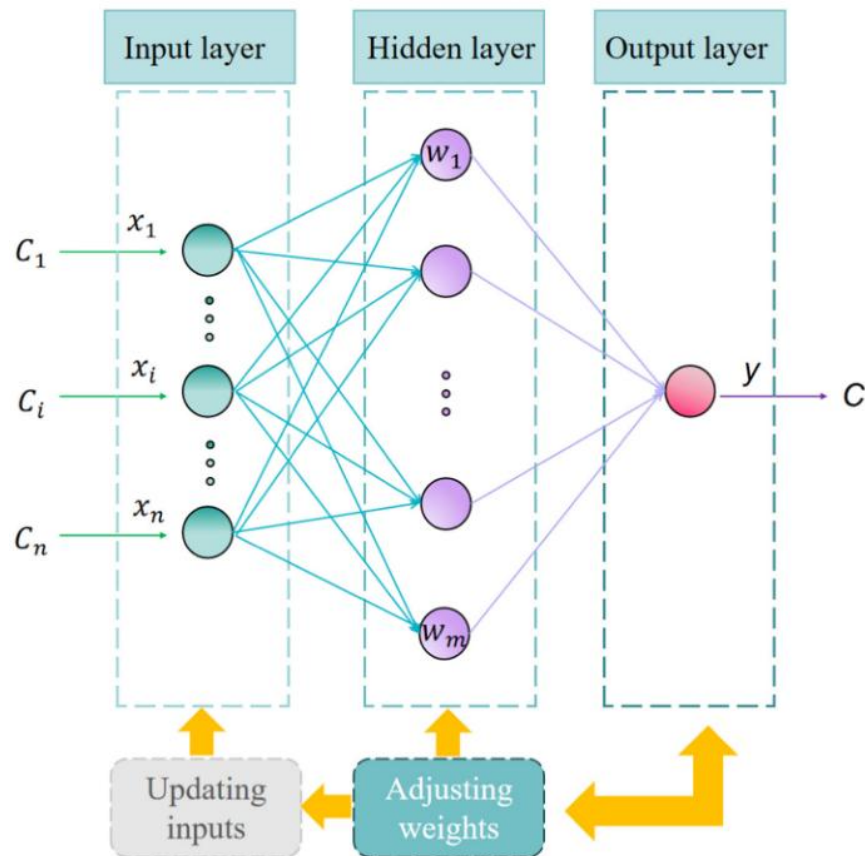


Fig. 8. The structure of NN.



SOH Estimation Models

Hybrid Methods

Hybrid SOH Estimation:

- Combines **Model-based + Data-driven** methods (e.g., Kalman Filter + Neural Network).
- **Why Hybrid?** It leverages the strengths of both worlds to improve accuracy and generalization.

Comparison Summary:

- **Filters:** Good for real-time, but **model-dependent**.
- **Machine Learning:** High accuracy, but requires massive training data ("Data Hungry").
- **Hybrid:** Highest accuracy and robustness, but computationally expensive (**Complex**).



Comparison of Models

Speed vs. Accuracy vs. Complexity

Statistical Filters (The "Fast" Option):

- **Strength:** Low computational complexity; excellent for **real-time BMS updates**.
- **Weakness:** **Highly sensitive** to measurement noise; struggles with non-linear systems.

Machine Learning (The "Precise" Option):

- **Strength:** High accuracy; strong generalization; effectively handles **non-linear data**.
- **Weakness:** **"Data hungry"** (needs large datasets); computationally expensive; lacks interpretability (Black Box).

Hybrid Methods (The "Balanced" Option):

- **Strength:** Combines the best of both worlds, **high accuracy with strong robustness**.
- **Weakness:** **Most complex** to implement; requires significant processing power.



Comparison of Models

Speed vs. Accuracy vs. Complexity

Table 2

Comparison of advantages and disadvantages of different SOH estimation methods.

Method	References	Advantages	Disadvantages
Statistical filter-based SOH estimation models	Kalman filter	Low computational complexity and fast model update speed	High requirement for measurement accuracy and not suitable for nonlinear systems.
	Particle filter	Wide applicability and high flexibility	Complex computation and reliance on resampling
Machine learning-based SOH estimation models	SVM	High accuracy and fast calculation speed	Difficulty in hyperparameter selection, issues with kernel function choice, and lack of output confidence intervals
	RVM	Easy parameter tuning and strong noise resistance	Slow computation speed and sensitivity to outliers
	GPR	High accuracy and output with probability distribution	High computational complexity and poor online update capability
	NN	High accuracy and easy modeling	High computational complexity, limited accuracy with small sample data, and inability to provide real-time estimation
	ELM	Fast computation speed, short processing time, and certain generalization capability	Low accuracy, no confidence interval output, and poor robustness.
SOH estimation based on hybrid methods	[105–109]	High accuracy and strong generalization capability	Complex computations



Comparison of Models

Estimation Error Comparison
(Based on Reviewed Papers)

Filter-based Performance:

- Error: **RMSE ~1.2% - 3.37%**.
- *Note:* Good for general use but precision varies with model quality.

Machine Learning Performance:

- **SVM/RVM:** High precision, often achieving **RMSE < 0.5%** (e.g., 0.357% in studies).
- **Neural Networks:** Robust performance, typically **RMSE ~1.0% - 2.0%**.

Hybrid Methods Performance:

- **Superior Accuracy:** Often achieves the lowest error rates.
- Reported **MAE as low as 0.014** and **RMSE ~0.032** in advanced implementations.



Estimation Error Comparison (Based on Reviewed Papers)

Comparison of Models

Table 3

The detailed synopsis of SOH prediction methods for battery packs.

Method	References	Health feature	Feature sources	Estimation error
Kalman filter	[71]	Internal resistance	Charging and discharging	RMSE = 1.2 %
	[72]	Maximum capacity and internal resistance of the battery cell	Charging and discharging	Error = 3.37 %
	[73]	Feature point voltage	Discharging	Error < 4.14 %
	[74]	Temperature, open circuit voltage, current	Charging and discharging	7.57 % < RMSE < 10.72 % (high current) 5.52 % < RMSE < 8.52 % (low current)
Particle filter	[76]	SOC	Charging and discharging	N/A
	[77]	Internal resistance	Charging	0.35 % < Variance < 0.46 %
	[78]	Internal resistance	Charging and discharging	0.75 % < RMSE < 1.16 %
SVM	[83]	Voltage and temperature profiles	Charging and discharging	RMSE = 0.357 %
	[84]	The peak values of IC curves	Charging	RMSE = 1.59 %
RVM	[85]	Partial charging curve	CC charging	Less than 3 % error at 95 %.
	[87]	Voltage and current data of battery pack	Partial DOD	MAE = 0.11 % RMSE = 0.17 % STD = 0.19 %
	[88]	Integral of charging voltage curve, integral of charging current curve, and integral of charging temperature curve within a specific voltage range	Charging	MAE = 0.31 % MAPE = 0.32 % RMSE = 0.42 %
	[89]	Equal voltage drop discharge Time	Discharging	RMSE = 1.18 %
GPR	[91]	Battery current, pack voltage, cell voltage, and cell temperature	Charging and discharging	RMSE = 2.19 % MaxAE = 5.64 %
	[92]	EIS	Charging and discharging	R ² value is generally > 0.85
	[93]	Random partial charge segments	Random charging data	MAE = 0.6 %



Estimation Error Comparison (Based on Reviewed Papers)

Comparison of Models

NN	[95]	Referenced-based Charging Time	Charging and discharging	RMSPE < 1 %
	[96]	Maximum voltage and minimum voltage of battery cell, Average voltage of battery pack, battery pack voltage, circuit current, average temperature of battery	Charging and discharging	Error = 2.02 %
	[97]	Voltage curves, current curves, charge capacity curves	Charging	RMSE = 1.503 %
	[98]	Discharge current, discharge time	Discharging	RMSE = 1.0 % MAEP = 0.4 %
ELM	[101]	The peak value of the imaginary part, the real part at the peak of the imaginary part, the amplitude at the peak of the imaginary part, the imaginary part and the amplitude at 0.1 Hz, the phase peak value	Charging and discharging	0.55 % < Error < 1.12 %
	[102]	Voltage difference between cells, charging Time difference, accumulated mileage, driver's charging behavior with temperature deviations, SOC at the end of discharge	Charging and discharging	RMSE = 0.0849 %
	[103]	The time it takes for the discharge voltage to drop from 3.8 V to 3.5 V, the average value of the battery voltage during discharge, the average value of the battery temperature during discharge, cycle number	Discharging	MAPE = 0.071 % MAE = 0.26 % RMSE = 0.33 %
	[104]	Increase in the ohmic internal resistance and the polarized internal resistance	Charging and discharging	Error < 2.5 %
Hybrid methods	[105]	Ohmic resistance, diffusion resistance	Charging and discharging	MAPE = 0.4165
	[106]	Accumulated mileage, statistical characteristic distribution of discharge multiplicity, ambient temperature and SOC usage frequency number	Charging and discharging	MAE = 0.014
	[107]	Peak point, intercept, and slope of IC curve	Charging and discharging	RMSE = 0.032 Error < 2 %
	[108]	The average and standard deviation of current, the sum and standard deviation of battery pack voltage, the standard deviation of SOC, the average of battery voltage difference, the maximum temperature of the battery, and the sum of temperature difference	Charging and discharging	MAE = 1.24 % RMSE = 1.53 %
	[109]	Capacity	Charging and discharging	0.7938 % < RMSE < 1.7768 % 0.0087 % < MAPE < 0.0217 %



Challenges & Future Prospects

Barriers to Real-World Implementation

The Measurement Problem:

- Accurate **internal parameters** (e.g., impedance) are difficult to isolate due to coupled electrochemical reactions and sensor limitations

The "Lab-to-Reality" Gap:

- **Training Data:** Models are often trained on clean laboratory data.
- **Real World:** Practical operation involves **noisy data**, **sensor errors**, and **poor data quality**, significantly reducing model accuracy .

Environmental Adaptability:

- Current methods struggle with **dynamic conditions** (fluctuating temperatures, varying discharge rates) common in EV driving, as **opposed** to stable lab environments .



Challenges & Future Prospects

The Road Ahead for SOH Technology

Adapting to New Tech:

- Developing models for **Solid-State Batteries** (the future of EVs), which have different electrolytes and safety profiles compared to current Li-ion batteries .

Advanced Sensing Technologies:

- Moving beyond Voltage/Current: **Integrating Ultrasonic, Laser, and Infrared Thermography** to "see" inside the battery without destroying it .

Smarter BMS Integration:

- **Goal:** **Embed Hybrid Models** directly into the Battery Management System (BMS).
- **Benefit:** **Enables real-time, adaptive management strategies** like optimized charging protocols to extend battery life .



Conclusion & Future Outlook

The Core Problem: Pack Inconsistency

- SOH estimation has shifted focus from single cells to Battery Packs.
- **Key Finding:** Cell inconsistency (the "Barrel Effect") is the primary driver of accelerated pack degradation, requiring specialized "Pack-Level" features..

The Best Methodology: Hybrid Models

- While Statistical Filters offer speed and Machine Learning offers precision, **Hybrid Methods** represent the optimal solution.
- They balance computational cost with high accuracy, minimizing errors better than single approaches.

The Goal: From Lab to BMS

- Current research succeeds in controlled environments but faces a **"Reality Gap"** (noise/temp).
- **Future Direction:** Integrating robust Hybrid models into real-time **BMS** to ensure safety and reliability in actual EV operations .



Thank you